00-MASTER-PLAN

LOT Pager — Master Plan (Hardware v0 “LOT-P1”)

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1. One-line product

A 40 × 40 mm stainless-steel pocket device (“LOT Pager”) that receives autonomous, AI-written notifications from lot-systems.com (“Coffee time!”), shows them on a small screen, senses local weather/air, sees through a camera, and sends a **COPY** button press back to the site’s **Log** tab.

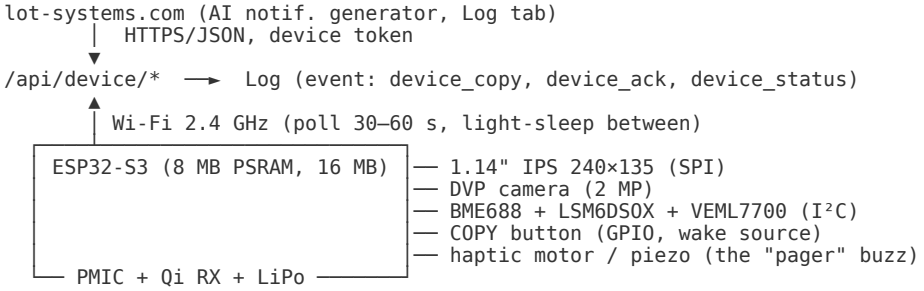
2. Requirement trace (your 19 points → where handled)

#	Requirement	Decision	Doc
1	PCBWay	PCB + PCBA + CNC stainless all quoted at PCBWay (single vendor for the 100 run)	07
2	Pager-like notification from AI site	Server writes `device_notification` rows; device polls/long-polls and buzzes + shows text	05
3	2-part stainless body	316L top shell + bottom shell, laser-welded or M1.2 screws + gasket	03
4	Flat silver square 4×4 cm × 5 mm	**Not achievable in full spec at 5 mm** (see §4). v0 = 40×40×9, v1 ≤ 7, 5 mm = stretch	03
5	Camera	2 MP DVP camera on ESP32-S3, stills only (QVGA/UXGA on demand)	01, 04
6	LOT API connector	Device token + `/api/device/*` endpoints (proposed, not yet built)	05
7	PDF manuals	Every doc here is rendered to `pdf/`	all
8	Compress info per session	`08-SESSION-DIGEST.md`, ≤1 page, rewritten each session	08
9	Firmware documents	Architecture, power states, OTA, boot flow	04
10	Software to connect with firmware	`lot-link` CLI/web-serial tool: flash, pair, test	06
11	Separate documents	8 standalone files, no cross-dependency to read one	—
12	Charger	Qi desk pad (off-the-shelf v0, custom TX v1)	01, 03
13	100 units	Pilot run plan, cost, yield, test	07
14	Weather sensor	Bosch BME688 (T/RH/P/gas)	01
15	AI-grade off-the-shelf sensors	BME688 (BSEC AI gas classes), LSM6DSOX (on-chip ML core), VEML7700 light	01
16	Button “Copy” → Log tab	`POST /api/device/event {type:"copy"}` → `Log` event `device_copy`	05
17	Polished stainless side	Back shell, mirror (Ra < 0.05 µm)	03
18	Other side: camera + screen + button	Front shell with glass window, camera hole, button	03
19	Wireless charger	Qi RX coil + ferrite behind a **non-metal** window	03

3. Two hard physics problems (decide before ordering anything)

1. **Stainless blocks Qi charging and Wi-Fi/BLE.** A fully metal sealed case = no wireless anything. Fix: (a) the seam between the two steel halves is a 0.8 mm PEEK/epoxy band that doubles as the antenna slot; (b) the Qi coil sits behind the glass/ceramic face, not behind steel. The device charges **face-down** on the pad.
2. **5 mm height.** Stack-up in `03-MECHANICAL.md` totals ≈ 8.7 mm for a rechargeable device with a real camera. 5 mm only works with a ≤1.2 mm battery (≈ 15 mAh = days→hours of life) and a wafer camera. Treat 5 mm as v2 research.

4. Architecture



5. Decisions needing the Inventor (blocking)

1. Accept 40×40×9 mm for v0? (Y/N)
2. Camera privacy: physical shutter/LED, or software-only? (Recommend LED + stills only.)
3. What does COPY copy? Recommend: the currently shown notification text → logged to the Log tab with timestamp.
4. Which LOT account owns a device — one device per user, token pairing from Profile page?
5. Budget cap for the 100 run (est. $17–26k, see 07).

6. Caveats on this document set

- vendor sites and lot-systems.com were **unreachable** from the build sandbox; part links are manufacturer/distributor search URLs and **prices are estimates — verify before ordering**.
- brand.lot-systems.com / institute CQGS pages were not readable; brand tokens were taken from `tailwind.config.js` (blue `#43aff3`, yellow `#fef17b`, black `#333`/`#050505`). Re-check against the brand guide.